A Dataset for Misbehavior Detection in Vehicular Named Data Networking Using the JubST SUMO Scenario

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***Abstract*—Vehicular Named Data Networking (VNDN) has emerged as a promising communication paradigm for intelligent transportation systems, offering improved scalability, caching efficiency, and content availability over traditional IP-based VANET architectures. However, these advantages also introduce new security vulnerabilities at the forwarding layer, where malicious vehicles can exploit name-based communication to disrupt data dissemination or manipulate routing behavior. Despite growing interest in VNDN security, the research community lacks a publicly available dataset that captures forwarding-layer misbehavior within a realistic VNDN architecture, making it difficult to evaluate, compare, and reproduce intrusion detection mechanisms.**

To address this gap, this paper presents a new dataset generated using the JubST SUMO mobility scenario integrated with VEINS and OMNeT++. The dataset includes four representative VNDN forwarding attacks—Interest Flooding, Interest Aggregation, Selective Forwarding (Gray Hole), and Name Prefix Hijacking (Route Hijack)—simulated under varying traffic densities, attacker ratios, and random seeds. A lightweight trust-based evaluation mechanism is incorporated to label malicious behavior and support supervised, unsupervised, and hybrid detection approaches. The final dataset comprises 225 simulation runs, providing detailed message logs, ground truth attacker annotations, and complete configuration files for full reproducibility.

This dataset establishes the first unified baseline for evaluating misbehavior detection mechanisms in VNDN, reducing simulation complexity, enabling fair comparison across studies, and supporting future extensions by the research community. It represents an essential step toward standardized benchmarking tools for securing next-generation vehicular communication networks.

***Index Terms*—Vehicular Named Data Networking, Cooperative Intelligent Transport Systems, intrusion detection system**

I. Introduction

VEHICULAR Ad Hoc Networks (VANETs) are a fundamental component of Intelligent Transportation Systems (ITS). They are specialized forms of Mobile Ad Hoc Networks (MANETs) that facilitate communication between vehicles and roadside infrastructure. In VANETs, vehicles are equipped with wireless communication devices that allow them to exchange information with other vehicles and fixed Roadside Units (RSUs). This network supports a variety of applications related to traffic management, road safety, entertainment, and internet access.

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VANETs utilize different levels of communication, including vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), and hybrid communication, which combines both V2V and V2I. They operate based on multiple communication standards, such as Dedicated Short-Range Communications (DSRC) and Cellular Vehicle-to-Everything (C-V2X). However, VANETs encounter several challenges, including high node mobility, frequent changes in network topology, broadcast storms, limited bandwidth, security threats, and the need to maintain quality of service (QoS).

Vehicular Named Data Networking (VNDN) has emerged as a promising paradigm for next-generation intelligent transportation systems, combining the content-centric communication model of Named Data Networking (NDN) with the dynamic topology and VANETs. By retrieving data based on names rather than host addresses, VNDN enhances scalability, caching efficiency, and content availability in vehicular environments. However, this architectural shift introduces a new range of security vulnerabilities that can be exploited by malicious or faulty vehicles, particularly at the forwarding layer where Interest and Data packets are exchanged dynamically. Such misbehavior can lead to serious degradation in network performance, congestion, and even compromise safety-critical applications such as collision warnings or cooperative awareness.

Despite significant progress in the design of VNDN-based intrusion detection systems (IDS), a major limitation persists in the form of **dataset availability**. Most prior studies rely on IP-based VANET datasets that do not accurately capture the name-based communication semantics or the hierarchical forwarding structures of VNDN. Consequently, the evaluation of VNDN security mechanisms often requires reimplementation of complex simulation environments, making fair comparison and reproducibility difficult. To address this gap, this paper introduces a new publicly available dataset. The dataset is generated using the *JubST* mobility scenario implemented in the SUMO simulation framework and modeled through the VEINS and OMNeT++ network simulators.

The proposed dataset incorporates four representative forwarding-layer attacks in VNDN: *Interest Flooding*, *Interest Aggregation*, *Selective Forwarding (Gray Hole)*, and *Name Prefix Hijacking (Route Hijack)*. Each attack is simulated under varying network conditions, attacker densities, and traffic loads to evaluate detection mechanisms in realistic vehicular environments. A lightweight trust-based evaluation mechanism is integrated to label and assess malicious forwarding behavior

in real time, enabling both supervised and semi-supervised learning approaches for misbehavior detection.

The main objective of this work is to provide the VNDN research community with a standardized, reproducible, and extensible dataset that serves as a baseline for evaluating misbehavior detection algorithms. This dataset not only reduces simulation complexity and setup time but also ensures consistency and comparability across different studies. Ultimately, the proposed dataset represents a foundational step toward establishing a comprehensive benchmarking framework for securing future VNDN-based vehicular communication systems.

I wish you the best of success.

mds
August 26, 2015

II. Related Work

Cooperative Intelligent Transport Systems (C-ITS) enable vehicles and roadside infrastructure to exchange information in real-time, improving traffic safety and efficiency. C-ITS applications, such as collision avoidance, congestion control, and cooperative perception, rely heavily on the timely dissemination of trustworthy data. However, the open and decentralized nature of VANETs exposes them to a broad range of security and privacy threats. Vehicles can be compromised, impersonated, or manipulated by different types of threats and attacks. Therefore, misbehavior detection systems (MDS)are essential for identifying malicious or faulty entities by analyzing behavioral, contextual, and communication features of vehicles. Traditional security mechanisms, such as Public Key Infrastructure (PKI) and certificate revocation lists (CRL), ensure authentication and message integrity, but cannot detect deceitful insiders, including legitimate participants who purposely misbehave. To design and evaluate effective MDS techniques, realistic datasets are essential. Prior research has initiated several datasets on misbehavior using the IP-based VANET environment. This section reviews the most relevant datasets designed for VANETs, specifically in an IP-basedVANET environment and emerging data-centric architectures.

The baseline benchmark that is used widely is the Vehicular Reference Misbehavior dataset (VeReMi) [1], developed to provide a standard testbed for evaluating position-based misbehavior detection mechanisms in vehicular environments. The data set was generated using the Luxembourg SUMO Traffic Scenario (LuST) [2] through the Simulation of Urban Mobility (SUMO) [3]and VEINS frameworks [4] integrated with the Misbehavior Detection Framework (F2MD) [5]. VeReMi includes five representative position-falsification attacks on Basic Safety Messages (BSMs) and follows a standardized logging format. VeReMi has been broadly adopted to validate both rule-based and machine-learning-based misbehavior detection models. Despite its influence, VeReMi includes only a limited number of attack types and omits environmental attributes such as traffic density, weather, or interference. Therefore, as an enhancement to the VeReMi baseline, the same authors proposed the VeReMi-Extension dataset [6] to enable more realistic, diversified, and comparable evaluation scenarios. This version introduced new attack categories and integrated sensor error models to simulate real-world measurement uncertainty in position, speed, acceleration, and heading. VeReMi-Extension dataset includes different misbehavior types, including message delay, data replay, Denial-of-Service (DoS), disruptive, and Sybil-based attacks. Although VeReMi-Extension increased dataset realism, it lacked temporal continuity between message sequences. The MisbehaviorX dataset [7] was developed using the VEINS framework integrated with the Vehicular Ad-hoc Simulation Platform (VASP) and modeled on realistic Boston-area traffic maps. It addresses the narrow attack coverage and limited scale of prior datasets by providing both benign and malicious traces encompassing 68 distinct types of misbehavior. The MisbehaviorX dataset encompasses various attacks, including position falsification, speed manipulation, data replay, DoS flooding, Sybil amplification, and collusion attacks, as well as hybrid behaviors executed by coordinated vehicle groups. MisbehaviorX records Cooperative Awareness Messages (CAMs) and Decentralized Environmental Notification Messages (DENMs) along with contextual features such as packet delivery ratio, received signal strength indicator (RSSI), delay, and mobility metrics. The VeReMi-AP dataset [8] extends the VeReMi framework to enable the evaluation of predictive and proactive detection mechanisms. VeReMi-AP introduces the concept of attack prediction, which estimates the possibility of upcoming malicious activity based on historical communication patterns. It also includes Fake Reporting Attack, where vehicles broadcast fabricated safety alerts (e.g., false accident or congestion notifications) that mislead other vehicles and degrade traffic flow. Temporal indicators such as message rate and inter-alert time are included to support time-series learning.

The BurST–Australian Dataset for Misbehaviour Analysis (BurST-ADMA) [9] was designed to model vehicular misbehavior within an Australian context using the Burwood SUMO Traffic (BurST) scenario from Melbourne. It features one-second interval Basic Safety Messages (BSMs) for urban and suburban environments, comprising both benign and malicious entries that inject false position, speed, and heading data. The dataset contains seven attack types, including constant and random position or speed falsification and reversed heading attacks. While BurST-ADMA diversifies the geographical scope and traffic patterns of vehicular datasets, it remains limited to BSM-level falsification without modeling network-level or content-centric misbehavior.

The reviewed datasets collectively contributed to advancing misbehavior detection research within VANETs. However, most of them are confined to IP-based architectures and focus primarily on physical or message-level falsification. The BurST–Australian Dataset focused on the data-centric environment. However, there are no currently available datasets that are suitable for VNDN-specific communication behaviors such as Interest/Data exchanges, cache manipulation, or privacy leakage through semantic naming. Moreover, none integrates multi-layer trust evaluation or temporal feature correlation. VeReMiVND uses a data-centric vehicular networking framework, enabling the study of sophisticated intrusion detection and secure forwarding strategies in VNDN.

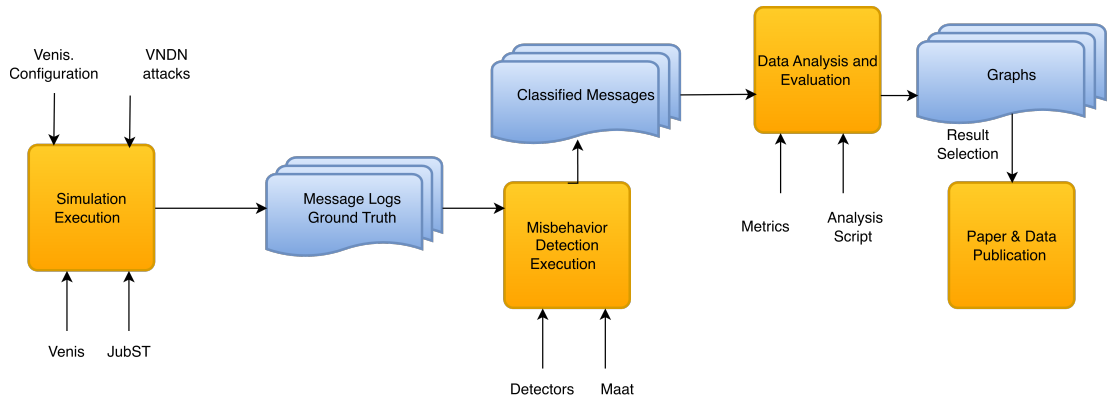

Fig. 1. Evaluation Overflow

III. JubST SCENARIO

A. Motivation

B. Overview of JubST Scenario

Jubail Industrial City is one of the largest industrial cities in the world, established in 1975. It is strategically located in Saudi Arabia, in the Middle East, covering an area of 1,016 square kilometers that includes various industrial complexes and port facilities. The city's proximity to energy sources and essential raw materials contributes to its significance. Jubail's advanced infrastructure has laid the foundation for industrial, commercial, and social sectors to thrive within an interconnected framework. It also meets transportation needs with modern standards, offering elements of practical daily life [10].

The JubST scenario was simulated using the Simulation of Urban Mobility (SUMO) [3] road traffic simulation suite. Below, the main steps involved in the simulation process are described:

1) Map Extraction: We utilized OSMWebWizard [11], available in the SUMO suite, for map generation and extraction for the Jubail City. The OSMWebWizard is a graphical user interface (GUI) tool that allows users to select an area for map and traffic generation. For this project, Jubail Industrial City was selected using a boundary box (bbox).
2) Traffic Demand Generation: The OSMWebWizard generates traffic based on the provided input. For this study, we focused solely on vehicles for simplicity, as our main interest is in generating different VNDN attacks. Traffic demand was generated randomly. Table I summarizes the inputs used for traffic demand generation. The simulation was run for an arbitrary time of —— seconds to generate a preliminary dataset. However, we plan to optimize the JubST scenario in the future to create a more realistic traffic demand and network topology for a longer duration.

C. Overview of VNDN Attacks

The VNDN architecture enhances the scalability, caching efficiency, and data availability of VANETs; however, it also introduces a new class of security vulnerabilities across multiple networking layers. The proposed dataset will focus on specific VNDN forwarding attacks/ architecture attacks. Instead of implementing a broad range of attacks, we are concentrating on these particular threats to demonstrate the efficacy of our approach. We anticipate that other researchers will contribute new attack implementations and corresponding datasets to our repository, which we will maintain.

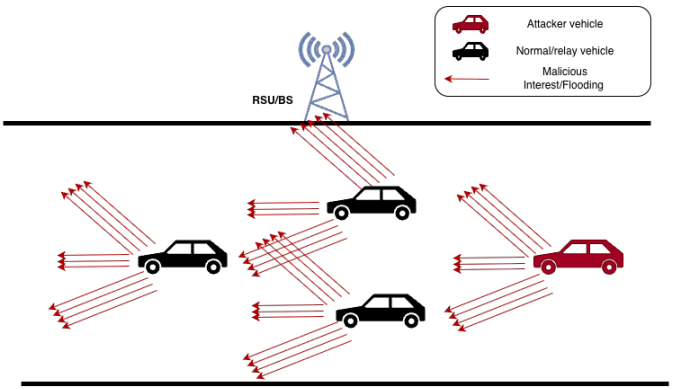

Fig. 2. A flooding Attack

Our approach involves testing some forwarding layer attacks of VNDN that can detect malicious forwarding behavior in real-time. We selected the following attacks: Interest Flooding, Interest Aggregation, Selective Forwarding (also known as Gray Hole), and Name Prefix Hijacking (Route Hijacking). Below is an overview of these VNDN attacks.

1) Interest Flooding (IF): IF attack is one of the most common DoS attack that targeted VNDN. In IF attack, a malicious vehicle generates a storm of Interest packets for non-existent, unpopular, or randomly varying content names. These interest packets must be recorded in the Pending Interest Table (PIT) of intermediate nodes, causing abnormal traffic leads to PIT overflow, queue congestion, and exhaustion of memory and bandwidth resources [12]. IF can affect the VNDN infrastructure even the other vehicles or RSUs. When multiple malicious vehicles broadcast a surge of Interests,the RSUs and forwarding nodes become overloaded, causing

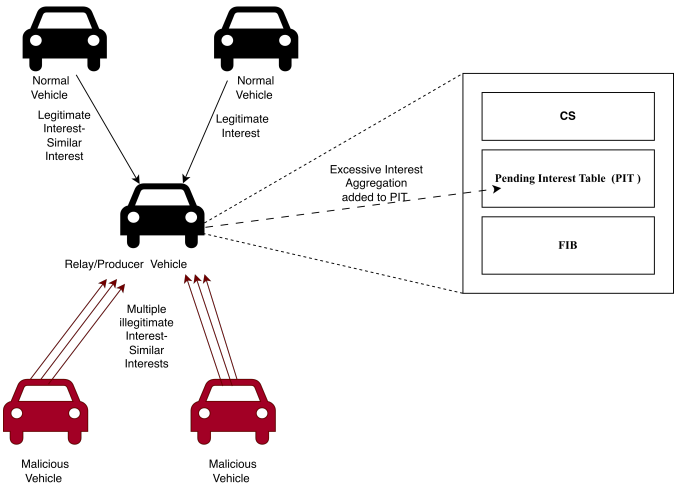

Fig. 3. Interest Aggregation in VNDN

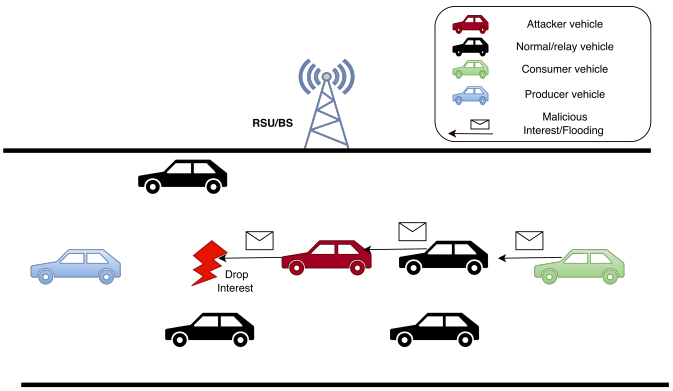

Fig. 4. Caption

legitimate Interest packets to be dropped or delayed, thereby disrupting real-time safety applications such as collision warnings, traffic congestion alerts, and emergency message dissemination. Over time, such attacks can destabilize the control plane of the vehicular infrastructure, impairing coordination between fog nodes, RSUs, and cloud controllers, which depend on consistent forwarding performance for route optimization and trust management [13]. Figure 5 depicts the IF attack in VDND.

2) Name Prefix Hijacking / Route Hijack: It is known as Prefix Hijacking or as Route Hijacking. It is a critical routing-layer attack in VNDN that targets the name-based forwarding mechanism.As a natural of NDN, Data packets are routed based on the name prefixes rather than IP addresses, and each node maintains a Forwarding Information Base (FIB) mapping content names to outgoing interfaces [12]. In this attack, a malicious vehicle falsely advertises ownership of certain name prefixes—for instance, /traffic/alert or /safety/data—and claims to be the legitimate source of that content. This deception causes nearby vehicles or RSUs to update their FIB entries and forward Interests toward the attacker instead of the genuine data producer. An attacker can exploit the dynamic environment of VNDN to intermittently hijack routes, redirect safety-critical data, or blackhole legitimate Interests, thereby interrupting time-sensitive applications such as collision avoidance and traffic coordination. Moreover, hijacked name prefixes can be used to inject false data (e.g., fake congestion warnings or bogus location updates). Figure 5 shows the scenario of this attack. In large-scale deployments, prefix hijacking can disrupt fog–cloud synchronization and cluster-based forwarding to alter the trust and reliability across the VNDN infrastructure.

3) Selective Forwarding (Gray Hole): The Selective Forwarding attack is a forwarding-layer misbehavior in which a malicious vehicle selectively drops Interest or Data packets rather than forwarding them normally. Unlike blackhole attackers, which discard all traffic, gray hole nodes forward a subset of packets to remain undetected, while strategically dropping those related to safety messages or trust-estimation data [12]. In VNDN, where timely content dissemination and multi-hop reliability are essential for maintaining vehicular safety and network stability, this attack reduces cache hit ratios, increases end-to-end delay, and disrupts cooperative forwarding, leading to reduced throughput and poor quality of service in fog-assisted vehicular networks. Furthermore, since the attacker still appears partially cooperative, trust-based detection models find it challenging to differentiate gray hole behavior from legitimate fluctuations in connectivity [14]. Recent studies have highlighted that such attacks can compromise cluster stability and RSU coordination, making the infrastructure vulnerable to cascading failures [15].

4) Interest Aggregation Attack: Interest aggregation is an NDN forwarding optimization that collapses multiple identical Interest packets arriving at a router into a single forwarded Interest and records the requesting interfaces in the PIT. While this reduces redundant traffic, it can be abused by adversaries. In an Interest Aggregation Attack, malicious nodes intentionally craft Interest patterns (e.g., repeated identical names, long lifetimes, or name variants that converge at a specific router) to force aggregation at strategic points in the network. By doing so the attacker can artificially inflate PIT occupancy at targeted routers, prolong PIT entries (by renewing or keeping Interests alive), and collapse distinct legitimate Interests behind a single aggregated Interest that may be delayed or lost. The outcome is increased response latency, content unavailability for legitimate users, degraded cache utilization, and a subtle DoS effect because multiple consumers' Interests are effectively serialized or starved. In vehicular environments (VNDN) these effects are magnified by high mobility and tight real-time requirements — safety messages or time-sensitive content may be delayed or suppressed when Interests are aggregated maliciously at overloaded Roadside Units (RSUs) or fog nodes, undermining both safety and QoS for applications such as collision warnings, cooperative

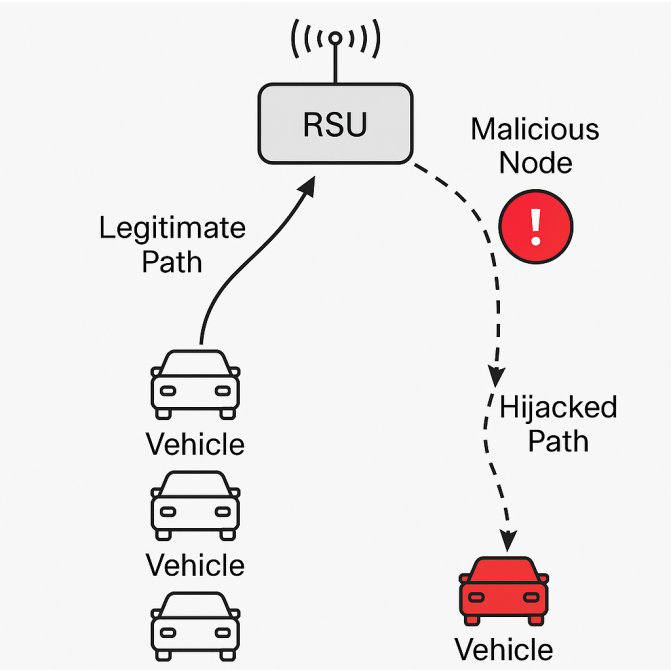

Fig. 5. A flooding Attack

perception, or traffic control.

IV. JubST-Dataset for Misbehaviour Analysis

V. Dataset Contribution and Motivation

The primary contribution of this paper is the introduction of a novel dataset specifically designed to fit within the VNDN architecture. The dataset aims to establish a common baseline for the evaluation of misbehavior detection mechanisms in VNDN environments. Meanwhile, previous studies have predominantly relied on IP-based vehicular datasets, such as those used in conventional VANET simulations, these datasets do not accurately represent the communication and forwarding mechanisms of VNDN. Although IP-based frameworks offer flexibility in defining custom attack scenarios, they fail to capture the unique characteristics of name-based communication, caching, and interest–data exchanges inherent to VNDN. To the best of our knowledge, there is currently no publicly available dataset that explicitly supports misbehavior detection research within the VNDN architecture.

The goal of this dataset is to provide an initial baseline against which detection mechanisms for VNDN can be systematically evaluated. This resource significantly reduces the time required for researchers to perform high-quality simulation studies and facilitates fair comparison across different detection frameworks. We acknowledge that no dataset can fully replace a detailed implementation and analysis of each detection mechanism. However, the current state of the art—where researchers must perform time-consuming and error-prone re-implementations of prior methods for each study—poses a major barrier to reproducibility and scientific progress. Our dataset therefore represents the first step toward a standardized and comprehensive evaluation methodology for misbehavior detection in VNDN.

The dataset comprises detailed message logs for every vehicle participating in the simulation, along with a ground truth file specifying the behavior of each attacker. Each log includes both network-level communication data and local information collected from the vehicle, such as periodic position updates from on-board sensors (e.g., GPS). This structure enables any detection algorithm to process the sequence of messages and assess the reliability or malicious intent of each event or packet exchange.

The dataset and all corresponding source code used to generate it are publicly available[1]. It currently consists of — encompassing diverse network conditions:

- ,
- attacker densities (low, medium, high),
- traffic densities, and
- repetitions per parameter set using different random seeds.

A detailed discussion of these parameters, the attack scenarios, and the data generation methodology is presented in the following subsections.

A. Scenario Selection

The purpose of the proposed dataset is to provide a holistic and unbiased foundation for the evaluation of misbehavior detection mechanisms within the VNDN architecture. The dataset is designed to minimize unintentional selection bias that may arise from the specific properties of a detection mechanism or simulation scenario, even if this means sacrificing some degree of detail in individual case studies. The common approach adopted in prior work has been to select a limited number of specific scenarios (e.g., a——-) to analyze the performance of a mechanism in depth. While this method provides valuable insights into behavior under well-controlled conditions, it relies heavily on manual decision-making and assumptions, thereby complicating fair and reproducible comparisons between different detection techniques.

In contrast, our work emphasizes how detection mechanisms perform across a diverse and representative set of scenarios. To achieve this, we constructed a larger, more comprehensive dataset that captures a variety of traffic and attacker configurations, allowing researchers to first assess overall detection performance before focusing on specific conditions for mechanism-specific improvements.

The dataset thus enables both broad generalization fine-grained analysis of detection schemes under different operational environments.

To ensure scenario diversity, we selected a representative sample of the complete simulation environment based on the included road types (urban and highway) and their corresponding traffic densities. Our simulation design was inspired by the JubST scenario, which was originally developed to provide a comprehensive and realistic testbed for evaluating VANET applications. As a result, most studies referencing prior VNDN work tend to rely on partial or follow-up experiments rather than full reproductions.

[1]Link to dataset repository will be provided upon publication.

This is precisely where our dataset makes a novel contribution. It provides a simplified and reusable message stream per vehicle, making it significantly easier to reproduce detection studies and to benchmark newly proposed mechanisms. The dataset preserves the essential temporal and behavioral features of each vehicle, including transmission events, Interest/Data exchanges, and local sensing data. This structure allows straightforward parsing by any detection algorithm while maintaining fidelity to the dynamics of vehicular communication.

Table I summarizes the core parameters of our simulation setup. Additional technical details, including complete configuration parameters, can be found in the OMNeT++ configuration files that are publicly available in our source repository.

B. VNDN Enabled Dataset Structure

C. Attacks and Implementation

D. Characteristics

E. Limitations

VI. Metrics

A. Evaluating Detection Quality

B. Evaluating Detector Limitations

VII. Conclusion and Discussion

Acknowledgment

The authors would like to thank...

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TABLE I
SIMULATION PARAMETERS FOR VNDN DATASET GENERATION

Parameter		
Mobility Model	SUMO JubST	Jubail Industrial City SUMO Traffic scenario used for realistic urban mobility [**?**].
Simulation Start Time		
Simulation Duration		
Attacker Probability		
Simulation Area		
Signal Interference Model		
Obstacle Shadowing Model		
Fading Model		
Shadowing Model		
MAC Implementation		
Thermal Noise		
Transmit Power		
Bit Rate		.
Receiver Sensitivity		
Antenna Model		
Beaconing Rate		

TABLE II
COMPARISON OF EXISTING VEHICULAR MISBEHAVIOR DATASETS AND THE PROPOSED VEREMIVNDN DATASET

Dataset	**Description and Simulation Environment**	**Attack Coverage and Features**	**Limitations / Missing Aspects**
VeReMi [1]	Baseline dataset for position-based misbehavior detection; generated using SUMO + VEINS + F2MD on the LuST scenario.	Five basic position-falsification attacks (*constant*, *random*, *constant-offset*, *random-offset*, *eventual-stop*).	Limited attack diversity; no trust, privacy, temporal, or environmental attributes; IP-based VANET only (no VNDN).
VeReMi-Extension [6]	Enhanced VeReMi using same framework with added sensor error models.	Adds new attacks: message delay, data replay, DoS, disruptive, Sybil-based; introduces measurement noise in position, speed, acceleration, and heading.	Still lacks temporal continuity, trust, and privacy indicators; IP-centric only.
VeReMi-AP [8]	Predictive extension of VeReMi for proactive detection; SUMO–VEINS with F2MD.	Introduces *Fake Reporting Attack* for forecasting misbehavior; includes temporal indicators (message rate, inter-alert time).	No trust or privacy layer; single-layer IP communication; excludes data-centric (VNDN) misbehavior.
MisbehaviorX [7]	Comprehensive dataset using VEINS + VASP over Boston maps.	68 attack types: position falsification, speed manipulation, replay, DoS, Sybil, collusion; includes temporal logs, RSSI, delay, and PDR metrics.	Large-scale but lacks trust, privacy, and VNDN-level semantics (Interest/Data, cache, or PIT features).
BurST-ADMA [9]	Australian dataset (Burwood SUMO Traffic scenario) with 1-s BSM intervals for urban/suburban settings.	7 attack types: position/speed falsification and reversed heading; supports local traffic diversity and left-hand drive models.	Confined to BSM-level falsification; no trust, privacy, temporal correlation, or network/content-centric features.
Proposed VeReMiVNDN	Next-generation dataset integrating VANET and VNDN simulation using SUMO + VEINS + NDNSim.	Multi-layer attacks (Interest flooding, content poisoning, cache pollution, Sybil amplification, replay); includes trust and privacy-aware features; temporal sequences; pseudonym rotation; privacy-preserving naming.	Designed to overcome previous gaps by combining temporal, trust, and privacy modeling in a data-centric vehicular environment.